

1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

Definition 1.10.1

The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$. The **vec operator** places the n columns of an $m \times n$ matrix one below another, giving a vector of length mn : $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$.

Proposition 1.10.1

For matrices of compatible sizes:

- (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ (mixed-product rule);
- (b) $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for nonsingular $\mathbf{A}, \mathbf{B}$. More generally $\mathbf{A}^- \otimes \mathbf{B}^-$ is a generalized inverse of $\mathbf{A} \otimes \mathbf{B}$;
- (c) $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A})\text{tr}(\mathbf{B})$ and $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A})\text{rank}(\mathbf{B})$;
- (d) if $\mathbf{A}$ ($m \times m$) and $\mathbf{B}$ ($p \times p$) are symmetric with eigenvalues λ_i and μ_j , the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are the products $\lambda_i\mu_j$, and $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p(\det \mathbf{B})^m$;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f) $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$ and $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$.

Proof.

- (a) The (i, j) block of the left side is $\sum_k a_{ik}c_{kj}\mathbf{B}\mathbf{D}$, which is the (i, j) block of the right side. (b) Transposition is direct, and the rest follows from (a), for example $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^-$.
- (b) With $\mathbf{A} = \mathbf{PAP}^\top$ and $\mathbf{B} = \mathbf{QMQ}^\top$, (a) and (b) give $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{A} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$, a spectral decomposition because $\mathbf{P} \otimes \mathbf{Q}$ is orthogonal and $\mathbf{A} \otimes \mathbf{M}$ is diagonal with entries $\lambda_i\mu_j$. The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ is idempotent with the same rank as $\mathbf{A} \otimes \mathbf{B}$ (Proposition 1.3.3(a)). Its trace is $\text{tr}(\mathbf{AA}^-)\text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A})\text{rank}(\mathbf{B})$ by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The j th column of $\mathbf{ABC}$ is $\mathbf{ABc}_j = \sum_k c_{kj}\mathbf{Ab}_k$, which is the j th block of $(\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$. The trace identity is $\sum_{ij} a_{ij}b_{ij}$ written two ways.

■

The determinant formula in (d) holds for all square $\mathbf{A}$ and $\mathbf{B}$, not only symmetric ones. We need only the symmetric case.

Example. (The balanced two-way layout)

Let observations y_{ij} , $i = 1, \dots, a$, $j = 1, \dots, b$, be stacked with j running inside i . The additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because $\mathbf{I}_a \otimes \mathbf{1}_b$ has a 1 in column i exactly for the b observations in row i . With $\bar{\mathbf{J}}_k = k^{-1}\mathbf{J}_k$, a symmetric idempotent averaging matrix, the mixed-product rule shows that $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$ and $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ replace each observation by its row mean, its column mean and the grand mean. The matrix $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ is symmetric and idempotent, with trace $a + b - 1 = \text{rank}(\mathbf{X})$. It is the projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 16.1.2 in Chapter 16). The script checks this for $a = 3$ and $b = 4$, together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")          # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))          # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),          # intercept
               np.kron(np.eye(a), one(b)),      # row indicators
               np.kron(one(a), np.eye(b))])     # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

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